A study of experimental privacy budget quantification in quantum differential privacy simulations

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ABSTRACT 150-250 palabras

INDEX TERMS Differential privacy, Quantum computing, Quantum noise

I. INTRODUCTION

ESTA va a ser la introduccion a la tesis. En esta sección pienso hablar de la motivación, la importancia de la privacidad diferencial, basicamente muy similar al documento de propuesta

II. BACKGROUND

En esta seccion se va a introducir todos los conceptos, ecuaciones y conocimiento necesario para entender el paper. Siguiendo el documento de propuesta propuesta vamos a tener:

A. QUANTUM INFORMATION AND COMPUTATION

Definition 2.1: Quantum states [1]: A quantum state of the system X is described by a density matrix ρ whose entries are complex numbers and whose indices (for both its rows and columns) have been place in correspondence with the classical state set Σ

Definition 2.2: Mixed state [2]: A mixed state of a quantum system is an ensemble

$$\mathbb{E} = \{(p_1, |\psi_1\rangle), ..., (p_k, |\psi_k\rangle)\}$$

which in turn can be represented by the density operator

$$\rho_{\mathbb{E}} = \sum_i p_i \left|\psi_i\right\rangle \left\langle\psi_i\right|$$

thus the resulting $\rho_{\mathbb{E}}$ will be the density matrix representation of the ensemble.

Definition 2.3: Quantum Channel [1]: A quantum channel describes a linear mapping over density matrices.

Quantum channels are important as they are the generalization of an operation over a quantum state and in the case of this project we need to use operations such as the depolarizing channel which is defined thanks to this general definition.

Definition 2.4: Bloch spehere [1]: The Bloch sphere is a useful geometrical representation for a single qubit.

Definition 2.5: Trace distance [2]: It is the distance between two density operators and it is defined by:

$$\tau(\rho, \sigma) = \frac{1}{2} Tr|\rho - \sigma|$$

where $Tr(x)$ is the trace of the density matrix x

another useful form to define the trace distance in terms of vectors by Nielsen & Chuang [3] is

Definition 2.6: Trace distance [3]: It is the distance between two qubit vectors u and v defined by:

$$\tau(u, v) = \frac{1}{2}\|u - v\|$$

Definition 2.7: Quantum measurement [2]: It is the way to extract information from a quantum system. Mathematically, a measurement is modelled as a set of matrices $M = \{K_m\}$ with the condition of

$$\sum_m K_m^\dagger K_m = I$$

with I being the identity of the density operator (i.e for a qubit $I = 2 \times 2$ identity matrix)

When talking about quantum differential privacy below it won't really matter what the quantum state collapses into. Given this liberty we can define a POVM (Positive Operator-Valued Measure) which is a simplification over the measurement where we get a set of matrices $M = \{M_m\}$ with $M_m = K_m^\dagger K_m \; \forall m$ this is possible since we only care about the probabilities and so $p_m = Tr(K_m \rho K_m^\dagger) = Tr(M_m \rho)$

Definition 2.8: Quantum noise [2]: It is an inherent physical process that occurs on the quantum state that causes it to change in some way.

For the purposes of this investigation we are interested in three types of noise: General amplitude damping, Phase-amplitude damping and Depolarizing noise. Each one of these types represent a unique physical process which is summarized in greater detail in [4] and [2]. However for this proposal it is only necessary to note that these processes will randomize the quantum state by a certain degree.

B. DIFFERENTIAL PRIVACY

Definition 2.9: (ϵ, δ)-Differential privacy [5]. A randomized algorithm $\mathcal{M}$ will provide (ϵ, δ)-Differential privacy (henceforth (ϵ, δ)-DP) if for any two outputs on two different databases D and D' differing by at most one record, and for all possible outputs A:

$$Pr[\mathcal{M}(D) \in A] \leq e^{\epsilon} Pr[\mathcal{M}(D') \in A] + \delta$$

C. QUANTUM DIFFERENTIAL PRIVACY

Definition 2.10: (Quantum differential privacy [2]). Let $d \in (0, 1]$ and $\epsilon, \delta > 0$ be three constants. A quantum operation $\mathcal{E}$ is (ϵ, δ)-differentially private if for every POVM $M = M_m$, for all outputs $S \subseteq Out(M)$ and for all inputs ρ, σ such that $\tau(\rho, \sigma) \leq d$, it holds that:

$$Pr[\mathcal{E}(\rho) \in_M S] \leq e^{\epsilon} Pr[\mathcal{E}(\sigma) \in_M S] + \delta$$

There are many ways to obtain that epsilon in theory however Zhou and Ying proposed the following algorithm to obtain a quantity called proportional distance which can help obtain ϵ.

Definition 2.11: (Proportional Distance [2]). For two density operators ρ and σ, their proportional distance is defined as:

$$PD(\rho, \sigma) = \sup_{\{M_m\}, m} \max \left(\ln \frac{q_m}{p_m}, \ln \frac{p_m}{q_m} \right)$$

where the supremum is over all POVMs $\{M_m\}$ and all possible outcomes m,

$$p_m \equiv Tr[\rho M_m], \quad q_m \equiv Tr[\sigma M_m]$$

are the probabilities of obtaining outcome m when the measurement is performed on ρ, σ, respectively, and we make the conventions $\frac{0}{0} = 1$, $\frac{1}{0} = +\infty$, $\ln 0 = -\infty$, $\ln +\infty = +\infty$.

III. RELATED WORK

It was important to figure out what the current investigative and academic situation was with regards to the fields of quantum information and computation and differential privacy. For this an investigation was made in which different databases were searched and analized to find out the amount of papers that covered the topics mentioned before. The investigation found 61 relevant papers that focused on the topics and the specific distributions of each one are compiled in Figure 1.

As shown in Figure 1 the investigation yielded positive results as there were many academic papers that not only mentioned the topic but gave valuable research and insight on possible angles the project could leap into. Among the most interesting ones are: [6] in which a hybrid quantum machine learning framework was proposed to allow differential privacy in the model. [4] which tested different quantum differential privacy mechanisms along with proposing its own **Lifted** quantum differential privacy, which enhanced the randomness induced in the quantum states. [7] in which an extended analysis was made to characterize the protection quantum noise can offer in the setting of quantum-classical algorithms. And finally in the study of Zhou and Ying [2] they define a framework of differential privacy in the quantum setting and examine how different noise mechanisms change the protection of a privacy budget ϵ.

IV. DATA PROCESSING AND ANALYSIS

This section details the data processing and analysis methodology that was used in order to prepare the classical Statlog (German Credit Data) dataset [8] [9] for use in the quantum setting. Firstly we outline the dataset characteristics, variable selection procedures and lastly quantum encoding procedures for the selected data.

A. DATASET DESCRIPTION

The selected dataset for this investigation is the Statlog (German Credit Data) dataset created by Professor Hans Hofmann [8]. This multivariate dataset aims to classify people as a good or bad credit risk. The orginal dataset is composed of 1000 instances that span 21 variables (7 numerical, 13 categorical, and the target).

In the original dataset created by Professor Hofmann, all non numerical variables use a attribute code to define their characteristics (e.g The status of a person whose checking account is less than 0 Deutsche Marks (DM) is A11) for that reason it was decided that the investigation was going to use a curated version of the dataset [9] which transforms all the labels to their definition as a means of simplifying the readers' understanding of the dataset by eliminating that intermediate layer of label and category. Every mention to the dataset henceforth will refer to this curated version and not the original compiled by Professor Hofmann

A simple data analysis concluded that the dataset didn't have any problem in terms of completeness (as all rows include all 21 variables) nor uniqueness (as there is no instance of an exact duplicate). For these reasons no special cleaning procedure was done prior to variable selection.

B. VARIABLE SELECTION WITH PHIK

In order to hold quantum data for a dataset that contains 1000 instances of 20 variables each, the simulation would, at a maximum, store 1000 $2^n \times 2^n$ density matrices, with n being the number of qubits used. This means that using the full 20 variables, one per qubit, would imply $2^{20} \times 2^{20} \approx 1.1 \times 10^{12}$ complex entries, and at 16 bytes each, that is about 16 TiB

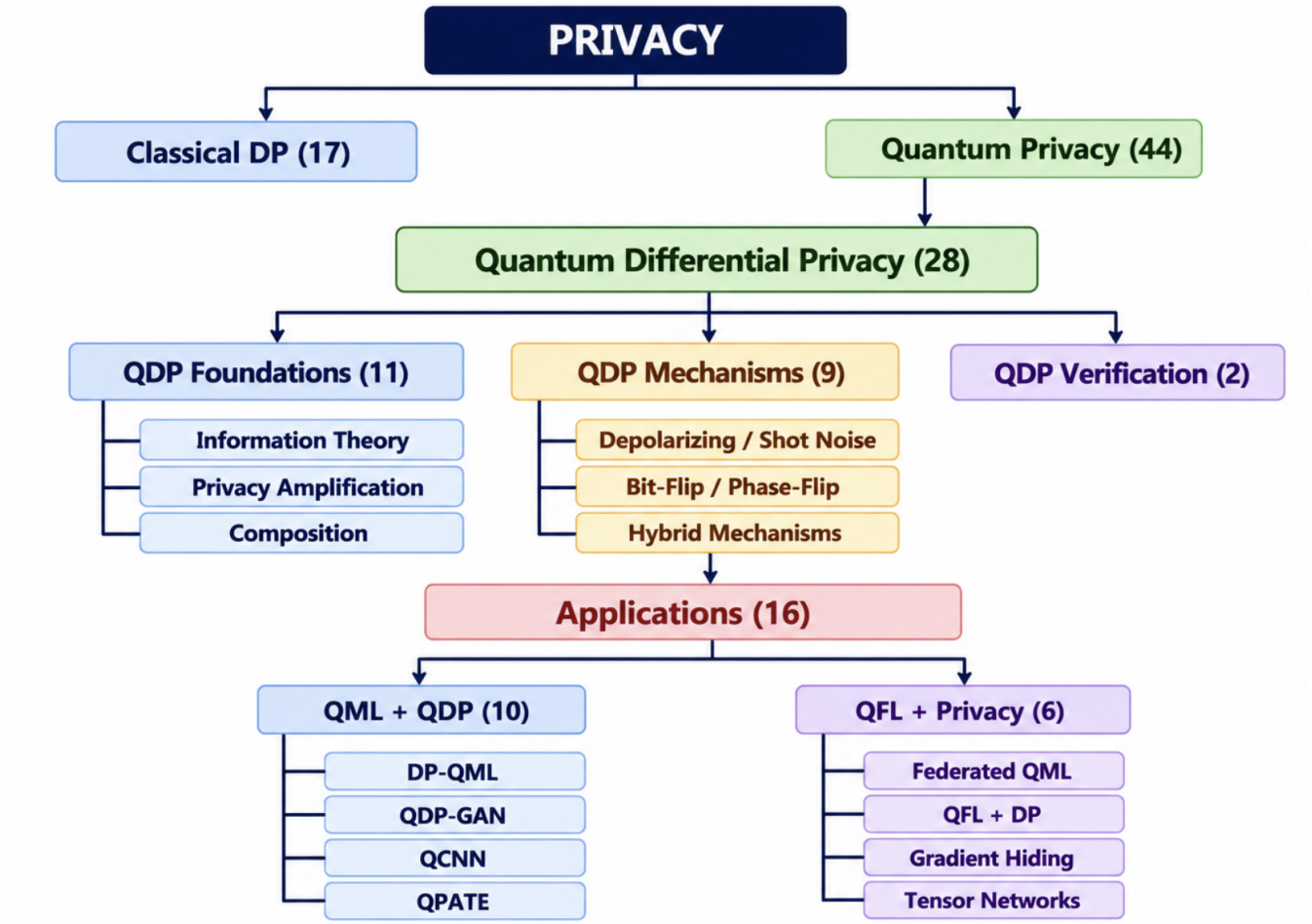


FIGURE 1. State of the art of quantum differential privacy. (own elaboration, based on the literature review described above)

per matrix. It was decided that a correlation study over the target variable would be an adequate way to reduce this load. The reasons for this decision are: First, the investigation did not aim to produce a classifier and thus could use a smaller number of variables in the experiments. Second, the variables selected from a correlation study would be the most likely candidates to be chosen as significant by a classifier.

As mentioned previously however, most of the variables in this dataset are not numeric and thus a popular correlation analysis such as the Pearson correlation is not an adequate choice to evaluate variable correlation. For starters, for a nominal variable the Pearson's coefficient relies on an arbitrary choice for the label and since the proportions of the target variable in the dataset for each category are not the same across the board, a change in category label implies a change in the coefficient. Moreover the Pearson coefficient identifies well the linear relationships of variables but lacks for nonlinear cases.

For these reasons it was decided to use the PhiK method [10] to select the variables which show a consistent stronger correlation with the target variable. PhiK is a stronger candidate as it works well for both numerical and categorical variables correlations and it is able to identify nonlinear relationships between them.

We used Baak et al. PhiK library to measure the correlation and significance between each variable and the target and recorded the results in Table 1. It is important to note that the library returns, for a pair of variables, two results: the first is ϕ_K which is the measurement of how strong the relationship between the two variables is. This result is calculated by calibrating a χ^2 test statistic to a contingency table (populated with the categorical variables or a bin distributed numerical variables) with respect to a bivariate normal distribution. This calibration returns a ϕ_K with range of 0 (no association) to 1 (complete association). Since ϕ_K depends on the counting of the contingency table renumbering the categories no longer affects the end result. The second result is Z which is a conversion of the p-value from the independence test into a number of standard deviations. A higher Z provides more evidence that the variable is not independent from the target.

For the variable selection we chose to use Z as although the ϕ_K measures the correlation between variables, the sig-

TABLE 1. ϕ_K correlation of each feature with the target and its statistical significance with a cutoff of $Z > 2.81$.

Feature	ϕ_K	Z	Significant
status_account	0.512	10.965	Yes
credit_history	0.197	6.912	Yes
month_duration	0.285	5.648	Yes
status_savings	0.147	5.283	Yes
credit_amount	0.241	4.388	Yes
collateral	0.217	4.009	Yes
purpose	0.204	3.715	Yes
housing	0.077	3.601	Yes
years_employment	0.098	3.053	Yes
other_installment_plans	0.063	2.891	Yes
is_foreign_worker	0.109	2.545	No
age	0.142	2.177	No
status_and_sex	0.123	1.953	No
secondary_obligor	0.041	1.835	No
payment_to_income_ratio	0.075	1.045	No
telephone	0.021	0.689	No
n_credits	0.000	0.043	No
job	0.000	-0.390	No
residence_since	0.000	-1.070	No
n_guarantors	0.000	$-\infty$	No

nificance Z of that correlation explains if that correlation is significant enough to be told apart from chance. The Bonferroni method states that, to ensure that the probability of at least one false positive across k tests (in our case $k = 20$ for 20 variables against the target) is at most $\alpha = 0.05$, each individual test must use a significance level of $\alpha' = \alpha/k = 0.05/20 = 0.0025$ [11]. Here a false positive is a variable that is independent of the target but is selected anyway. Since $\alpha' = 0.0025$ corresponds to $Z \approx 2.81$, we used $Z > 2.81$ as the cutoff for the selection of the variables.

As the reader can observe in Table 1 the selection took 10 out of the 20 variables which as mentioned before now leaves a maximum of 1000 $2^{10} \times 2^{10}$ matrices as opposed to 1000 $2^{20} \times 2^{20}$ should we use one qubit per variable, we will show how we lowered that number in Section IV-C. The definition and type of each variable is recorded in Appendix A

C. ENCODING OF CLASSICAL VARIABLES INTO QUANTUM

Having chosen a reduced set of variables from the dataset (10 from the original 20), the next step was to encode the classical information from each variable into qubits which could be used in the quantum simulation. For this it is important to note that the variables were divided into two groups each with their own encoding:

1) Categorical Variables

These are categorical variables whose categories do not follow a natural ranking. An example for this is the variable `purpose` which has ten categories (car(new), vacation, education, among others). As the reader can observe these categories don't follow a natural order as there is no way to tell if say car goes after or before education. Due to this it was decided to separate them from the categorical (ordinal) variablese that will be mentioned later on

The proposed encoding for these types of variables is given by Definition 4.1

Definition 4.1: (Method for encoding categorical variables) For a categorical variable with K categories. When ecoding the label $c \in \{0, \ldots, K-1\}$ the resulting encoded qubit state $|\psi\rangle$ is given by the equation

$$|\psi\rangle = RZ(\phi)H\,|0\rangle \text{ with, } \phi = \frac{2\pi c}{K}$$

This encoding ends up with a qubit state in one of K equally distributed points across the equator of the Bloch sphere. We choose to place the qubit on the equator as because from Definition 2.6 we can express τ in terms of the angle between to points of a circle Δ and the radius of that circle as $r \sin \frac{\Delta}{2}$. With this expression the reader can note that the minimum distance τ is maximized for all other angles on the sphere when the radius is maximum which happens in the equator of the sphere. More over one could argue that having a smaller θ (i.e the angle of rotation with regards to the y-axis of the Bloch sphere) would reduce ϵ, and one would be right if not for the fact that it comes with the tradeoff of making the categories less distinguishable. In the case where $\theta = 0$ any rotation over the z-axis would invalidate since no rotation over the y-axis would make it a global phase.

This method assures then that τ between two categories is at least $\sin\left(\frac{\pi}{K}\right)$. What this means in theory is that the privacy in the local setting won't change the final ϵ as regardless of the labels given to any category, the angle between the labels themselves might change but the possible angles between different categories will remain the same. However, what this encoding doesn't prevent is the fact that different categories can have different proportions. The labels given to each category can impact the distance τ between the dataset and that same dataset without a person included, which, as will be explained in Section V is crucial when computing ϵ_{exp} and ϵ_{theo} and then in theory can lead to different results. In Appendix B we compiled the c value given to the variable's categories.

2) Numerical and ordinal variables

These are variables which are either ranges of an amount (ordinal) or numerical. An example of these variables is `month_duration` which explains the loan's duration in months. A loan that goes for 4 months is clearly smaller than that of a loan that goes for 72, thus it is important to preserve that order. The encoding method which we believed would be able to preserve this order was Dense angle encoding [12].

The question that remained however was, which variables were going to be encoded together? To answer this question we first created all 15 possible arrangements of the 5 numerical and ordinal variables into two pairs and a single variable. Then, we once again used the PhiK method to analyze the correlation between each pair of variables which is documented in Annex 4. From this small investigation it was concluded that the combination of variables that yielded the smallest correlation in the

worst case was `status_account` as the lone variable `status_savings` and `month_duration` as the first pair and `years_employment` and `credit_amount` as the second pair. We chose this combination as the smallest correlation between the worst pair would imply that the qubits carried as little redundant information within each qubit as possible.

Before the encoding definition we defined a function **minmax**(a, min, max) which normalizes a value $a \in [min, max]$ to $a \in [0, \pi]$

Definition 4.2: Given a value $a \in [min, max]$ with $min, max \in \mathbb{R}$ we define the function **minmax** as

$$minmax(min, max, a) = \pi \times \frac{a - min}{max - min}$$

With this definition the method to encode numerical and ordinal variables is Dense Angle Encoding [12] which we define as:

Definition 4.3: (Method for encoding numerical and ordinal variables with dense angle encoding). For an ordinal variable with K categories and a numerical variable with range $[n_{Min}, n_{Max}]$. When encoding the ordinal variable with label $c \in \{0, \ldots, K-1\}$ and a numerical variable with value $n \in [n_{Min}, n_{Max}]$. The resulting dense angle encoding is given by

$$\begin{aligned} |\psi\rangle &= RZ(\phi)RY(\theta)\,|0\rangle\,, \\ \text{with } \phi &= minmax(n_{Min}, n_{Max}, n), \\ \text{and } \theta &= minmax(-0.5, K - 0.5, c) \end{aligned}$$

Because of pairing selection the definition only needs to cover the case for when an ordinal variable is paired with a numerical one. Thus it is important to note that the reason **minmax** is calibrated from -0.5 to $K - 0.5$ for the ordinal variables is so that the resulting categories cannot reach the poles which would result in the gate RZ being invalidated. This calibration then sets each category into the middle of a bin of length $\frac{\pi}{K}$. An exception to this definition is the lone variable `status_account` since it doesn't need this calibration as there is no RZ gate to invalidate so the range for **minmax** is $[0, K - 1]$

As the reader can observe thanks to dense angle encoding we can actually reduce the number of qubits necessary for encoding the full 10 variables to 8 which helps reduce the computational toll from the original $2^{10} \times 2^{10}$ matrices with 1:1 qubit to variables encoding to $2^8 \times 2^8$ matrices. The final quantum circuit can be seen on Figure 2

D. NOISE PARAMETERS SELECTION FOR EACH NOISE MECHANISM

Un pequeño parrafo que sustente porque se utilizaron los parametros de ruido dado (i.e p = 0.5, gamma = 0.5 etc) esto basicamente hace referencia al paper de Zhou.

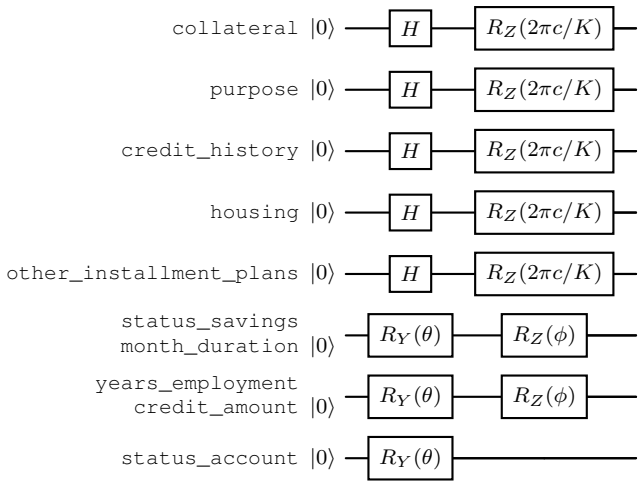


FIGURE 2. 8 qubit encoding circuit for the 10 selected variables

V. FINDING THE EXPERIMENTAL PRIVACY BUDGET

A. CENTRAL SETTING

La explicacion del experimento central, es basicamente puesto en texto la presentacion de semana 8

B. LOCAL SETTING

La explicacion del experimento loca, es basicamente puesto en texto la presentacion de semana 10

VI. MEASUREMENT AND CLASSICAL RELEASE OF CATEGORICAL VARIABLES

La explicación de la medicion de las variables categoricas como forma de mostrar el ruido en acción. Es el resto de la presentación de semana 10 puesto en texto

VII. EXPERIMENTAL RESULTS

A. FINDING THE EXPERIMENTAL PRIVACY BUDGET

1) Central Setting

Resultados del experimento central

2) Local Setting

Resultados del experimento local

B. MEASUREMENT AND CLASSICAL RELEASE OF CATEGORICAL VARIABLES

Resultados de las mediciones con las distribuciones de probabilidad

VIII. DISCUSSION AND CONCLUSION

En esta seccion voy a hablar de limitaciones encontradas dentro de los experimentos.

Entre ellas:

La dualidad entre C y L para la privacidad. (pros y contras) Que limitaciones tiene el experimento (analisis conjunto)

Tambien el analisis de los resultados de los mismos y que implicaciones tienen estos frente a la privacidad diferencial tanto clasica como cuantica.

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APPENDIX A DESCRIPTION OF SELECTED FEATURES

Table 2 compiles the name, type and description that were selected to use in the quantum experiments.

APPENDIX B DESCRIPTION OF LABEL ASSIGNED TO EACH CATEGORICAL VARIABLE

Table 3 compiles labels given to each variable using the scikit-learn python library.

APPENDIX C ϕ_K CORRELATION FOR ALL COMBINATIONS OF NUMERICAL AND ORDINAL VARIABLES

Table 4 shows the results for all the ϕ_K correlation between all combinations of numerical and ordinal variables, as well as their worst correlation between the two pairs.

TABLE 2. Description of the selected variables.

Name	Type	Description
status_account	Categorical (Ordinal)	Status of the existing checking account, in DM (Deutsche Marks)
month_duration	Numerical	Duration of the credit in months
credit_amount	Numerical	Credit amount
collateral	Categorical	Possible collateral for the loan
purpose	Categorical	Purpose of the credit
credit_history	Categorical	Credit history of the applicant
housing	Categorical	Current housing situation (rented, owned, or free)
years_employment	Categorical (Ordinal)	Length of time in current employment
status_savings	Categorical (Ordinal)	Status of the savings account, in DM (Deutsche Marks)
other_installment_plans	Categorical	Whether the client has other installment plans

TABLE 3. Assignment of the label given to each categorical variable's category

Variable	Category	Label
collateral	car	0
	none	1
	real estate	2
	savings agreement/life insurance	3
purpose	business	0
	car (new)	1
	car (used)	2
	domestic appliances	3
	education	4
	furniture/equipment	5
	others	6
	radio/television	7
	repairs	8
	retraining	9
credit_history	all credits at this bank paid back duly	0
	critical account/ other credits existing (not at this bank)	1
	delay in paying off in the past	2
	existing credits paid back duly till now	3
	no credits taken/ all credits paid back duly	4
housing	for free	0
	own	1
	rent	2
other_installment_plans	bank	0
	none	1
	stores	2

TABLE 4. Comparison of Variable Pairs and PhiK Metrics

			ϕ_K		
Alone	First pair	Second pair	First pair	Second pair	Worst
status_account	status_savings, month_duration	years_employment, credit_amount	0.000	0.043	0.043
credit_amount	status_account, years_employment	status_savings, month_duration	0.088	0.000	0.088
status_savings	status_account, month_duration	years_employment, credit_amount	0.091	0.043	0.091
status_account	status_savings, credit_amount	years_employment, month_duration	0.092	0.065	0.092
years_employment	status_account, month_duration	status_savings, credit_amount	0.091	0.092	0.092
month_duration	status_account, years_employment	status_savings, credit_amount	0.088	0.092	0.092
status_savings	status_account, credit_amount	years_employment, month_duration	0.114	0.065	0.114
years_employment	status_account, credit_amount	status_savings, month_duration	0.114	0.000	0.114
credit_amount	status_account, month_duration	status_savings, years_employment	0.091	0.150	0.150
month_duration	status_account, credit_amount	status_savings, years_employment	0.114	0.150	0.150
credit_amount	status_account, status_savings	years_employment, month_duration	0.200	0.065	0.200
month_duration	status_account, status_savings	years_employment, credit_amount	0.200	0.043	0.200
status_account	status_savings, years_employment	credit_amount, month_duration	0.150	0.689	0.689
status_savings	status_account, years_employment	credit_amount, month_duration	0.088	0.689	0.689
years_employment	status_account, status_savings	credit_amount, month_duration	0.200	0.689	0.689

ACKNOWLEDGMENTS

Esta seccion solo para la tesis o no se si pueda ir en el paper tambien

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